Fourier Series of an even functoin $f(x)$

$f(x) = \sum_{n=0}^{\infty} a_n \cos(nx)$ To find the coefficients a_n,

multiply both sides by $\cos(n'x)$ where n' represents any integer.

$$f(x)\cos(n'x) = \sum_{n=0}^{\infty} a_n \cos(nx)\cos(n'x)$$

Integrate both sides over the period of the functoin

$$\int_0^{2\pi} f(x)\cos(n'x)\,dx = \sum_{n=0}^{\infty} a_n \int_0^{2\pi} \cos(nx)\cos(n'x)\,dx$$

Because cosine is an orthogonal function the integral on the right hand side is zero except when $n=n'$.

If $n'=0$ then when $n=n'=0$

$$\int_0^{2\pi} \cancelto{1}{\cos(0\cdot x)}\,\cancelto{1}{\cos(0\cdot x)}\,dx = 2\pi$$

so

$$a_0 = \frac{1}{2\pi}\int_0^{2\pi} f(x)\,dx$$

otherwise if $n'=0$ an $n \neq 0$

$$\int_0^{2\pi} f(x)\,dx = \sum_{n=1}^{\infty} a_n \underbrace{\int_0^{2\pi} \cos(nx)\,dx}_{=0} = 0$$

So when $n'=0$, all terms in the series are zero except $n=0$.

If n' is some number other than zero and $n=n'$

$$\int_0^{2\pi} f(x)\cos(n'x)\,dx = a_{n'}\int_0^{2\pi}\cos(n'x)\cos(n'x)\,dx = a_{n'}\int_0^{2\pi}\cos^2(n'x)\,dx$$

$$= a_{n'}\int_0^{2\pi}\frac{1+\cos(2n'x)}{2}\,dx$$

$$= a_{n'}\Big[\underbrace{\frac{1}{2}\int_0^{2\pi}dx}_{=\pi} + \underbrace{\frac{1}{2}\int_0^{2\pi}\cos(2n'x)\,dx}_{=0}\Big]$$

$$= a_{n'}\,\pi$$

so

$$a_{n'} = a_n = \frac{1}{\pi}\int_0^{2\pi} f(x)\cos(nx)\,dx$$

All terms in the series are zero except $n=n'$

$$\therefore\ f(x) = \sum_{n=0}^{\infty} a_n \cos(n\pi)$$

$$a_n = \begin{cases} \frac{1}{2\pi}\int_0^{2\pi} f(x)\,dx, & n=0 \\ \frac{1}{\pi}\int_0^{2\pi} f(x)\cos(nx)\,dx, & n\neq 0 \end{cases}$$

Side note: Because $f(x)$ is an even function

$$\int_0^{2\pi} f(x)\cos(nx)\,dx = 2\int_0^{\pi} f(x)\cos(nx)\,dx$$

$$\Rightarrow\ a_n = \begin{cases} \frac{1}{\pi}\int_0^{\pi} f(x)\,dx, & n=0 \\ \frac{2}{\pi}\int_0^{\pi} f(x)\cos(nx)\,dx, & n\neq 0 \end{cases}$$